

Starting life

Most animals start life as a fertilized egg made up of a single cell. Eggs are often microscopic and not all of them have shells, but from this simple beginning the original cell divides many times. Gradually, a new individual called an embryo starts to take shape. In mammals, embryos usually develop inside the mother's body until the young are ready to be born. For many other animals, life really gets underway with hatching, when a new animal emerges from its egg into the outside world.

Tear creates opening in soft shell



Snake first uses its egg tooth to cut several slices in the shell



Yolk sac supplies developing shark with nutrients

Growth sac

Many sharks and rays lay pouch-shaped eggs with tough, fibrous cases. This egg was laid by a Japanese swell shark that lives near coasts in the Pacific Ocean. The case has tendrils at its corners, which anchor the egg to seaweed while the fish embryo develops inside, taking nourishment from the yolk sac.

Lone breakout

Some snakes give birth to live young, but rat snakes lay eggs with leathery shells. About 10 weeks after the eggs are laid, the young snakes hatch. They use a pointed scale on the tip of their upper jaw called an egg tooth to slice their way through the shell. Unlike mammals or birds, these reptiles are not looked after by their parents. From the moment they hatch, the rat snakes have to fend for themselves.



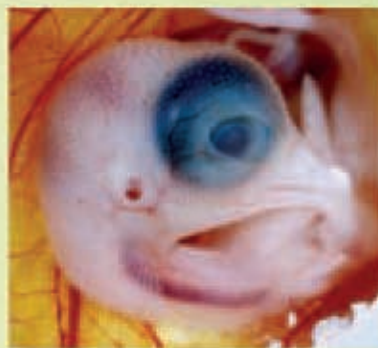
How embryos develop

A chick starts life even before the egg is laid by the hen. The single cell divides to form a hollow disk lying on the yolk sac. In the days after the egg is laid, the disk starts to form different organs and the chick's heart begins to beat. The embryo develops rapidly, using up the food stored in the yolk. After three weeks, the chick is ready to hatch.



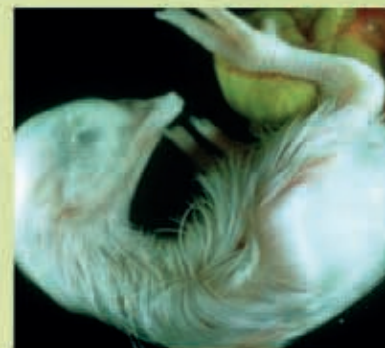
Microscopic embryo

Day 3—the head, eyes, and leg buds form. The heart pumps blood containing food and oxygen to the developing cells.



Growing body

Day 9—the embryo is still featherless, but has a large head and legs. The yolk sac shrinks as the chick's body grows.



Fully formed chick

Day 19—with just 48 hours to go until hatching, the chick now has feathers, wings, and well-developed legs.